HERDER
ELEKTRONISCHE SYSTEMEN

EMBEDDED SYSTEMS, CONTROLS AND TEST DEVELOPMENT

Hardware, firmware, *made to work as one.*

Herder Elektronische Systemen helps recover, build and validate electronic systems across hardware, firmware, controls and instrumentation.

DISCUSS YOUR PROJECT →

EXPLORE OUR EXPERTISE ↓

SERVING CLIENTS INTERNATIONALLY EMBEDDED SYSTEMS, CONTROLS AND ELECTRONICS

SCROLL TO EXPLORE ↓

Engineering status

IN REVIEW

01 WHAT WE DO

Engineering across the interfaces *where prototypes usually fail.*

The difficult part is often the space between disciplines. We bring hardware, firmware and control together, then test the system as a whole.

01

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Embedded controls and firmware

We design firmware around the hardware it has to control, then document the interfaces, states and tests needed to move the prototype forward.

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Power electronics prototyping

We help shape converter prototypes, bring them up on the bench and use measurements to guide the next design decision.

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Test automation and validation

We build repeatable tests that connect instruments, data and analysis, so the results are easier to trust and review.

04

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System integration and technical recovery

When a project has inherited code or unclear behavior, we trace the problem across hardware, firmware and communications and leave a documented path forward.

02 OUR APPROACH

Make the critical decision *while it is still cheap to change.*

We use requirements, prototypes and measurements to expose the important tradeoffs before they become expensive failures.

BRING US YOUR CHALLENGE →

A

Understand

We start with the system, its users and its constraints.

B

Engineer

We test the assumptions that matter, integrate the pieces and resolve the problems that block progress.

C

Enable

You leave with a working system, the evidence behind it and documentation another engineer can use.

03 SELECTED WORK

Representative *technical projects.*

These projects show how the work looks in practice: working hardware, firmware, controls and validation. They are public technical projects, not completed client contracts.

EMBEDDED CONTROLS 01

ESP32 Robotics Control Platform

A complete robotics-control platform integrating inertial sensing, wireless operator input, encoder feedback, motor drive, servo control and self-balancing PID behavior.

ESP32 / C++ | IMU / PID | Encoders | MATLAB / Simulink

VIEW PROJECT → VIEW REPOSITORY ↗

POWER ELECTRONICS 02

5V / 1A SMPS

An offline flyback prototype documented from transformer winding and component selection through waveform capture and bench validation.

Flyback SMPS | TNY285 | Waveforms | Validation

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SAFETY-ORIENTED CONTROLS 03

TI C2000 Actuator Controller

A hardware-backed F28069M demonstrator for command handling, explicit state control, PWM output, setpoint validation and fault response.

TI C2000 | SCI / UART | ePWM | Python tools

VIEW PROJECT → VIEW REPOSITORY ↗

04 FEATURED PUBLIC WORK

Controls at *testbed scale.*

PUBLIC UNIVERSITY RESEARCH AND ENGINEERING WORK

TI Piccolo dual-motor PMSM dynamometer

A dual-motor dynamometer built to develop, exercise and validate PMSM control across embedded firmware, CAN communications, real-time testing and measured torque-speed data.

Jonathan’s contribution

Jonathan integrated the dual-motor control system, migrated telemetry from SCI to CAN, connected the testbed to Speedgoat real-time hardware and developed validation workflows for speed, torque, power and efficiency data.

Evidence and methods

The repository includes architecture documentation, hardware integration notes, MATLAB/Python processing, efficiency maps and measured speed, torque and power data.

Publication boundary

This work was completed in a university laboratory setting. It is not a private client engagement or an exclusively owned product. The site links to the public repository without reproducing its models or large source files.

VIEW PROJECT → VIEW REPOSITORY ↗

Built for the *whole picture.*

01 **System architecture**

From requirements to clear interfaces

02 **Analog & digital electronics**

Design support for prototypes

03 **Industrial connectivity**

Communication that fits the system

04 **Verification & validation**

Results tied to test conditions

06 — CREDIBILITY BY PROCESS

Architecture → Prototype → Integration → Validation

The process keeps risk visible and gives each phase a clear next step.

07 INDEPENDENT PRACTICE

Engineering with *direct ownership.*

Herder Elektronische Systemen is an independent engineering practice led by Jonathan, focused on embedded systems, controls, electronics, test development and system integration.

The practice is intentionally small and hands-on, rather than a large staff or production organization.

START A TECHNICAL DISCUSSION

Let’s make it *real.*

Send a short brief. We’ll ask the right questions, clarify the next step and help define what the work needs.

- Project type and current stage

- Technical problem, target schedule and expected deliverables
- Approximate budget range and preferred contact method
- If files are needed, we will arrange secure exchange after the initial discussion

Project details are handled confidentially. Please do not send sensitive files in the initial email; secure file exchange can be arranged afterward. Public portfolio material is published only when authorized.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com

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